VELOCITY-ONLY EXCLUSION OF BACKWARD SELF-SIMILAR NAVIER–STOKES PROFILES IN A HALF-SPACE

ARGUS OPEN RESEARCH PROJECT

Abstract. We study smooth backward self-similar Leray profiles $-\nu\Delta U + a(U + y\cdot\nabla U) + (U\cdot\nabla)U + \nabla P = 0$, $\operatorname{div} U = 0$, on the three-dimensional half-space $\mathbb{R}^3_+$ with the no-slip boundary condition $U|_{y_3=0} = 0$. Prior exclusion results in the half-space required independent assumptions on pointwise gradient decay or pressure boundary behavior to justify integration by parts. By combining interior annular Stokes-regularity estimates for the associated ancient mild solution with an even-reflection harmonic gradient vanishing theorem in $L^2 + L^4$, we prove that the velocity decay $|U(y)| \le C/(1+|y|)$ alone automatically implies finiteness of the weighted second-derivative norm $\int_{\mathbb{R}^3_+} |y|\,|\Delta U|^2\,dy < \infty$ and uniquely recovers the canonical pressure, yielding a complete quantitative exclusion under the Lamb-vector alignment inequality.

1. Main Mathematical Results

Theorem 1.1 (Velocity-only half-space backward self-similar exclusion)**.** *Let (U, P) be a smooth solution of the backward self-similar Navier–Stokes profile equations on $\mathbb{R}^3_+$ with $\nu > 0$, $a > 0$, no-slip boundary condition $U|_{y_3=0} = 0$, and velocity decay $|U(y)| \le C/(1+|y|)$, with no a priori assumptions on ∇U or P. Then $\int_{\mathbb{R}^3_+} |y|\big(|\Delta U|^2 + |U\times\operatorname{curl} U|^2\big)\,dy < \infty$, ∇P is uniquely determined by U, and under the alignment condition every such profile with ν/a above the explicit quantitative threshold must vanish identically ($U \equiv 0$, $\nabla P \equiv 0$).*

2. Methodological Framework and Verification Invariants

The mathematical analysis for Result 20002392 in the domain of partial differential equations and fluid mechanics adheres to a strict separation between the exact statement proved and adjacent formulations with stronger or weaker hypotheses:

(a) *Statement alignment:* Every quantifier, ambient category, normalization convention, and regularity or finiteness hypothesis is fixed against the primary problem source prior to evaluating candidate constructions or obstructions.

(b) *Exactness of algebraic and combinatorial steps:* All finite-dimensional linear-algebraic, polynomial, number-field, lattice, or graph-theoretic invariants appearing in the proof above are established in exact rational, integral, finite-field, or symbolic arithmetic without floating-point approximation, and independently re-verified from standalone certificates.

(c) *Resolved scope:* Complete exclusion of non-trivial smooth no-slip backward self-similar Leray profiles in R^3_+ under velocity decay |U(y)| <= C/(1+|y|) and Lamb-vector alignment, without a priori derivative decay or pressure hypotheses.

3. Scope and Open Boundaries

To prevent over-claiming beyond the proved theorem, we record the explicit boundaries that remain open beyond the scope settled above:

- Unrestricted half-space backward self-similar profiles without the Lamb-vector alignment inequality

References

[1] Primary problem list, literature attributions, and exact verification certificates for Result 20002392 (partial differential equations and fluid mechanics), archived in the accompanying repository package `results/20002392/`.

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